

Arnav Mahadev

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EDUCATION

University of California, Berkeley

Bachelor of Science in Electrical Engineering and Computer Sciences - GPA: 3.62/4.0

Berkeley, CA

Expected May 2028

RELEVANT COURSEWORK

Machine Learning, Data Structures, Efficient Algorithms, Database Systems, Computer Security, Computer Architecture, Internet Architecture, Discrete Math, Probability Theory, Linear Algebra, Multivariable Calculus, Circuits & Devices

EXPERIENCE

Maadhev LLC

Software Engineering Intern

May 2026 – Present

Cupertino, CA

- Built a custom ACH payment system (Python, Flask, Stripe API) integrated with a Shopify app, enabling bank transfer payments at **70%** lower per-transaction cost than card processing.
- Engineered Stripe webhook handlers to sync payment and fulfillment data between Stripe and Shopify in real time, automating order confirmation across **50+ transactions**.
- Rebuilt the Shopify storefront and checkout flow, consolidating a multi step checkout into a single page.

iPick.ai

Software Engineering Intern

Jan 2026 – May 2026

Berkeley, CA

- Architected a graph-based company relationship visualizer using React, D3.js, and PostgreSQL.
- Built a supplier/subsidiary relationship extraction pipeline across **5,000+ public company tickers**, storing structured supplier/subsidiary edges in PostgreSQL to power graph-based visualization in the frontend.
- Integrated the Anthropic API to generate real-time news summaries, enriching company profiles with live information.

Claythis

Software Engineering Intern

May 2025 – Aug 2025

San Mateo, CA

- Created a Unity-based game in C# showcasing Claythis's AI-driven 3D model generation pipeline, implementing core weapon & health systems such as reloading, recoil, animations, armor, and healing.
- Tested and validated integration of the AI-generated 3D character models into Unity, verifying rigging, animation, and rendering behavior across the pipeline's outputs.

PROJECTS

Falsetto | Python, PyTorch, Hugging Face Transformers, scikit-learn, Gradio, Modal, librosa

Jul 2026 – Present

- Developed an AI-generated music detector using MERT embeddings and an attention-based AudioCAT classifier.
- Built an end-to-end ML pipeline achieving **87.5% accuracy, 0.972 AUC**, and validated with **56 automated tests**.

DEFCON | Python, PyTorch Geometric, scikit-learn, XGBoost, pandas, Plotly, NumPy

Jul 2026 – Present

- Built a graph attention network assigning defensive credit across **286K tracking frames and 2,100+ actions**.
- Beat naive positional baselines by **25%** on ranking accuracy and cut pass-outcome error by **67%** over a base model.

SoccerBoard | Python, JavaScript, XGBoost, PyTorch, FastAPI, Docker

May 2026 – Jul 2026

- Built a match predictor backtested on **2,500+ matches**, improving outcome accuracy from **47.5% to 59.9%**.
- Developed an expected goals model matching StatsBomb's professional benchmark **within 3.2%** of actual goals.

SaveBites | JavaScript, Python, React, HTML/CSS

Sep 2025 – Dec 2025

- Built the grocery-tracking backend for a food-waste app that scans receipts to auto-populate ingredient inventory.
- Developed a RESTful API for full CRUD on MongoDB ingredient data, fed by an OCR receipt pipeline.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C#, SQL

Backend: Node.js, Express, Flask, FastAPI, REST

Frontend: React, HTML/CSS, Liquid

ML/Data: XGBoost, PyTorch, scikit-learn, NumPy, pandas

Cloud/DevOps: AWS (EC2, S3), Vercel, Fly.io, Docker

Tools: Git, pytest, JUnit

ACTIVITIES

Eagle Scout | Scouts BSA Troop 476

Mar 2017 – Feb 2024

- Planned and led a classroom painting project, managing 15 volunteers end-to-end.
- Mentored 5 scouts and completed 150+ service hours, 75+ nights camping, and 120+ miles backpacking.

Grassroots Referee | U.S. Soccer Federation

Jun 2016 – Jun 2024

- Officiated 300+ youth soccer matches, managing conflict among players and coaches to ensure safety and fair play.
- Recognized with the 2017 Youth Referee of the Year Award by the San Jose Soccer Referee Association.